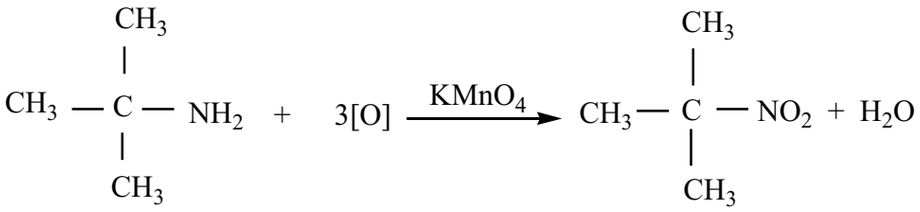
**Evaluate yourself**

**4) Find out the product of the following reactions.**

i) $CH_3CH(Cl)COOH \xrightarrow[ii)\ H_2O/\Delta]{i)\ NaNO_2} ?[X]$ ii) $CH_3CH_2\text{-}Br + NaNO_2 \xrightarrow{\text{alcohol }/\Delta} [Y]$

### 4) Oxidation of tert – alkyl amines

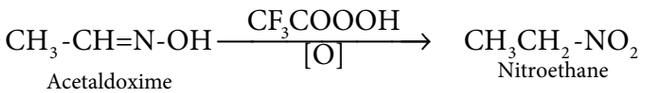
tert – butyl amine is oxidised with aqueous $KMnO_4$ to give tert – nitro alkanes.



tert-butylamine — 2 - methyl - 2 - nitro propane

### 5) Oxidation of Oximes

Oxidation of acetaldoxime and acetoneoxime with trifluoroperoxy acetic acid gives nitroethane ($1^0$) and 2 – nitropropane ($2^0$) respectively.

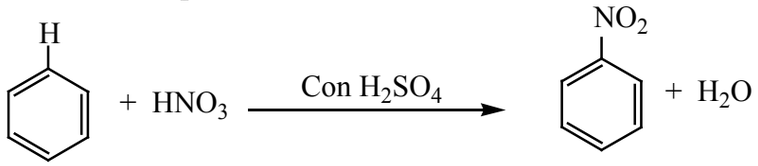


Acetaldoxime — Nitroethane

## 13.1.6 Preparation of Nitroarenes

### 1) By Direct nitration

When benzene is heated at 330K with a nitrating mixture $(Con.HNO_3 + Con.H_2SO_4)$, electrophilic substitution takes place to form nitro benzene. (Oil of mirbane)

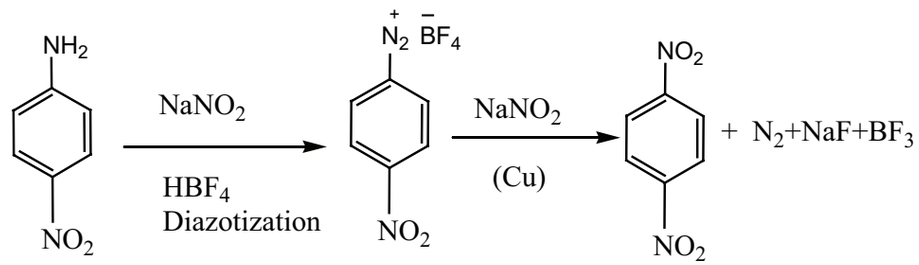


On direct nitration of nitrobenzene m- dinitrobenzene is obtained

### 2) Indirect method

Nitration of nitro benzene gives m-dinitrobenzene. The following method is adopted for the preparation of p-dinitrobenzene.

**For example**



p - nitroaniline — p - nitrodiazo fluoroborate — p - dinitro benzene



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